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Office building energy consumption forecast: Adaptive long short term memory networks driven by improved beluga whale optimization algorithm

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ABSTRACT

With the development of urbanization, buildings have become a major source of energy consumption. This research uses a data-driven approach to achieve accurate building energy consumption prediction by analyzing and modeling building energy consumption data. The model proposed in this paper uses improved beluga whale optimization algorithm (IBWO) to optimize long short-term memory networks (LSTM) for accurate energy consumption prediction. In order to enhance the ability of BWO in global search and local exploitation, a new method of dynamic adjustment of step factor as well as strategies such as nonlinear decreasing are introduced to improve BWO. For the first time, it is proposed to explore the accuracy of the number of hyper-parameters of LSTM on the prediction of energy consumption, and the improved beluga whale optimization algorithm is used to optimize the two, three, and four hyper-parameters of LSTM respectively. Then short-term prediction of historical energy consumption data of an office building in Xi'a is performed. Experiments show that the optimization of the four hyper-parameters of LSTM using the IBWO of this paper can reduce the mean absolute error (MAE) of the pre-improvement model from 830.71 KW to 128.28 KW, the mean absolute percentage error (MAPE) from 12.32 % to 1.38 %, and the coefficient of variation (CV) from 7.5 % to 1.2 %.

1. Introduction

With the continuous development of smart grid, renewable energy, and energy storage technologies, the need for energy consumption prediction is becoming more and more urgent. Traditional energy consumption prediction methods are usually based on historical data and statistical models, but these methods often fail to capture complex energy consumption patterns and dynamic changes. In order to improve the accuracy and precision of energy consumption prediction, in recent years, researchers have started to adopt meta-heuristic algorithms to solve the energy consumption prediction problem. Meta-heuristic algorithms are able to better adapt to changes and uncertainties in energy consumption patterns by simulating nature or social behavior in search and optimization.

1.1. The necessity of forecasting building energy consumption

By accurately forecasting building energy consumption, potential energy wastage problems can be identified promptly and relevant

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2352-7102/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Table 1
Recent studies.

Author	Technique	Dataset	Metric
Inna Bilous, Valerii Deshko and Iryna Sukhodub,2018 [9]	the paper is to develop a convenient tool building energy performance analysis based on regression model for internal air temperature prediction.	Verification of the received nonlinear regression model is performed based on simulation data using January climatic data from the IWEC file	R^2 Fisher's criterion
Tae Young Kim and Sung Bae Cho, 2019 [7]	Convolutional neural network and long short term memory	Individual household electric power consumption dataset from UCI machine learning repository	MSE, RMSE, MAE, and MAPE
Nivethitha Somu and Gauthama Raman M R, 2020 [10]	The improved sine cosine optimization algorithm is used to optimize the hyperparameters of LSTM	Case study on the real-time energy consumption data obtained from Kanwal Rekhi building, an academic building at Indian Institute of Technology, Bombay for short, mid, and long-term forecasting.	MAE MAPE MSE RMSE Theil statistics.
Aleksandar S. Anđelković and Dušan Bajatović,2020 [11]	An hourly city-scale Adaptive Neuro-Fuzzy Inference System algorithm with the Gaussian membership function is developed for both heating and non-heating regimes.	the City of Novi Sad of the Republic of Serbia is selected. According to the data from the Historical Archives of the City of Novi Sad.	R^2 MAPE
Deyslen Mariano-Hernández and Luis Hernández-Callejo, 2021 [12]	Different machine(MLR, ANN,RF,XGBoost, LSTM,TCN) learning and deep learning models were used to analyze which could perform better with the proposed strategy.	The presented forecasting strategy is tested using real data from two buildings located in Valladolid, Spain.(Energy consumption, climatic data)	R^2 RMSE MAE MAPE
X.J. Luo and Lukumon O. Oyedele, 2021 [13]	Genetic algorithm (GA) and long-short term memory (LSTM) neural network model for accurate and robust energy prediction.	Case study: Q block and S block of the Frenchay campus of University of the West of England, Bristol(electricity consumption, weather data)	MAE RMSE R^2
Anam-Nawaz Khan , Naeem Iqbal and Atif Rizwan, 2021 [14]	Spatial and temporal ensemble forecasting model: Long Short-Term Memory Unit (LSTM) and Gated Recurrent Unit.	A residential building in Seoul, South Korea (Hourly Electric Consumption data)	MAE, RMSE, MAPE R2 Score
Tomasz Cholewa, Alicja Siuta-Olcha, and Andrzej Smolarz,2021 [15]	A prediction method of thermal power demand for space heating is proposed, The method takes into account both meteorological parameters (external factors) and occupant behavior (internal factors).	The research for the purposes of this study was carried out in two buildings, a multi-family residential building (B1) and a school (B2) located in Poland.	MAPE, Outdoor temperature error
Tomasz Cholewa, Alicja Siuta-Olcha, and Andrzej Smolarz,2021 [16]	Presents a new simplified method. The method requires as input only the actual heat supplied to the heating system and the local outdoor weather conditions of a building. The output is a building energy model in terms of an equivalent outdoor temperature.	Analysis of detailed monitoring data of the heat transported by the three buildings (A,B,C) and the prevailing weather parameters (outdoor temperature, wind speed, and solar exposure) at their sites.	R^2 MBE CVRMSE
Zengxi Feng and Maoqiang Zhang, 2022 [17]	A dynamic combination residual forecasting model (FM) with the optimal combination approach: the SVR residual errors are forecasted by the dynamic combination FM based on the weights, and the SVR forecast value is finally corrected.	Case study is from an office building in Xi'an, a western city(air conditioning power, motive power, special power and lighting socket power consumption., outdoor meteorological information)	Relative errors, MAE, MAPE RMSE
Hangfu Wei and Xiaoqing Bai,2022 [18]	the integration of SSA, a CNN, and a BiGRU neural network	the energy consumption database of office buildings in the UK(electricity and gas consumption).	MAE, RMSE MAPE
Wenqiang Jing and Meng Zhen,2022 [19]	● Chaos theory ● the back-propagation (BP) neural network	Energy consumption data for a mall(Water consumption, Electricity consumption, Gas consumption and Central heating/cooling capacity)	RMSE and MAPE
Qiming Fu, Ke Li and Jianping Chen,2022 [20]	●the deep deterministic policy gradient (DDPG) method ●a deep-forest-based DQN (DF–DQN) method	The data set concerned was the energy consumption data in an office building in Shanghai.	MAE MAPE, RMSE R^2
Shuo-Yan Chou and Anindhita Dewabharata,2022 [21]	●the empirical mode decomposition ●wavelet transformation ●the long short term memory algorithm(LSTM)	The dataset from Taiwan consists of four buildings data.(residential in the University, laboratories, classrooms and offices)	MAPE, MAE MSE
Ruochen Lei and Jian Yin,2022 [22]	●Levenberg–Marquardts algorithm ●BP neural network	A super high-rise building located in the financial district of Yujiapu in Tianjin.(Cold and heat load)	MAE RMSE
Junhui Huang and Sakdirat Kaewunruen,2023 [23]	●establish a transformer-based model ●Support Vector Regression(SVR)	Main library at the University of Birmingham.	CV R^2 RMSE
Dongsu Kim, Yongjun Lee and Kyungil Chin,2023 [24]	●long-term short-term memory (LSTM) ●transfer learning (TL) strategies	A prototype hospital building model was built using Energy Plus to create input and test datasets for the LSTM-TL implementation	MBE, RMSE CV R^2
Peijun Zheng, Heng Zhou and Jiang Liu,2023 [25]	●Daily energy consumption pattern recognition (DECPR) method ●Temporal fusion transformers (TFT) model	Utilizes local building energy consumption data located in Zhenjiang city, Jiangsu province, China.(Hourly energy consumption, meteorological data)	MAE, MAPE, RMSE R^2

energy conservation measures can be developed. In addition, building energy consumption prediction can help optimize energy scheduling and management, and provide sensible energy consumption recommendations to reduce energy costs and minimize environmental impacts. However, accurately predicting building energy consumption remains a challenging problem as building energy consumption is affected by a combination of factors, such as building design, climatic conditions, activity patterns, and user behavior. Currently, building energy consumption prediction is becoming more and more sophisticated with the continuous development of data processing technology, sensor technology, and machine learning. Accurate building energy consumption prediction can be performed by using sensor collection data and historical energy consumption data, combined with mature data analysis and modeling techniques. Energy consumption forecasting can help enterprises, organizations, and individuals to rationally plan and manage energy resources in order to reduce energy waste and improve energy efficiency. Therefore, energy consumption prediction has an important application value in the energy field.

1.2. Existing approaches of building energy prediction

Currently, there are three main categories of building energy consumption prediction models in researching the progress of high prediction accuracy and high robustness: (1) engineering or white-box methods (EnergyPlus, eQuest, Ecotect, etc.), (2) statistical or grey-box methods (temperature frequency method, degree-day method, residential load factor method, etc.), (3) data-driven or black-box methods (neural networks, support vector machines, decision trees, regression models, k-nearest neighbor, etc.) [1,2]. Among them, data-driven methods have been widely used in the field of building energy prediction due to their accuracy, refinement, real-time, and adaptability.

A neural network is a computational model that mimics the nervous system of the human brain. It consists of a large number of interconnected artificial neurons (also known as nodes or neurons) that transmit and process information through weights. Neural networks are widely used in the field of machine learning and artificial intelligence to perform tasks such as pattern recognition, data classification, prediction, and decision-making. The basic unit of a neural network is the neuron. Each neuron receives inputs from other neurons and weights sums the inputs through an activation function and passes the result to the neurons in the next layer. This transfer process can be seen as the flow of information through the network. A neural network usually consists of multiple layers, including an input layer, a hidden layer, and an output layer. The signal enters the network from the input layer, passes through the hidden layer by layer, and finally gets the output result. However, most of the building's historical energy consumption data are closely related to time series, yet traditional neural networks are less effective in processing long time series data. Therefore, the Long Short-Term Memory (LSTM) network is proposed, and by introducing the gating mechanism, the long-term dependency can be effectively captured, thus improving the prediction accuracy of the model [3].

1.3. Existing approaches for hyper-parameters tuning

Existing hyperparameter tuning methods are generally divided into two types: manual search and automatic search. Manual search is mainly based on experienced or expert tuning. Therefore, it has disadvantages such as time-consuming and labor-intensive for non-expert users. As the number of hyperparameters introduced by neural networks increases and the range of values increases, it becomes more difficult to deal with hyperparameters manually [4].

In order to overcome the drawbacks of manual search which is time-consuming and subjective, automatic search algorithms are proposed. Optimization of hyperparameters of neural networks is a black box problem, so traditional optimization techniques such as gradient descent are not available. The existing automatic search algorithms are grid search, Bayesian optimization, and swarm intelligence algorithms. Lattice search [5] is a very simple method. It can exhaustively search in a given hyperparameter space and find the optimal combination of hyperparameters. However, the disadvantage of lattice search is the high computational complexity, especially when the number of hyperparameters is large or the range of values is large, the search space grows exponentially, resulting in a long search time.

Bayesian optimization has been proposed to address the drawbacks of grid search. Bayesian optimization uses a priori information and Bayesian inference to construct a probabilistic model and evaluates it by continually selecting those combinations of hyperparameters that promise better performance. Bayesian optimization can find better hyperparameter settings in fewer iterations and performs well in high-dimensional, continuous hyperparameter spaces. However, the general function of Bayesian optimization obeys a Gaussian distribution. Evolutionary algorithms simulate the process of biological evolution, searching for optimal solutions through crossover and variation among individuals in a population. When applied to hyperparameter optimization, evolutionary algorithms can gradually optimize the hyperparameters by generating new combinations of hyperparameters through continuous iteration and selecting and evolving them according to their performance. It is generally used to estimate the hyperparameters of neural networks.

1.4. A data-driven approach to forecasting building energy consumption

Table 1 presents recent research on data-driven building energy consumption prediction models. The literature review shows that the most common prediction method in the field of building energy consumption prediction is optimizing neural networks through evolutionary algorithms. The main types of neural networks are convolutional neural networks, artificial neural networks, and recurrent neural networks [6]. Regardless of which neural network is used for energy consumption prediction, the accurate selection of hyperparameters is very significant for the accuracy of the prediction [7]. However, as neural networks become more and more complex and the number and range of their hyperparameters increase, the accurate selection of hyperparameters for the prediction model becomes more and more important. To solve this problem, an automated search is needed to determine the optimal hyperparameters [8]. Therefore, the use of swarm intelligent optimization algorithms to determine the hyperparameters is a commonly used method to achieve higher prediction accuracy.

1.5. Why long short term memory networks?

Currently, in the field of building energy consumption prediction, common prediction models include artificial neural network models as well as time series prediction models. Statistical regression models predict future energy consumption by fitting the relationship between historical energy consumption data and relevant influencing factors. A time series model is a model that is specifically designed to deal with time-dependent data. Among them, the Autoregressive Moving Average Model (ARIMA) and Seasonal Autoregressive Moving Average Model (SARIMA) are common time series models [26]. Among these models, Recurrent Neural Networks (RNN) have been widely used in nonlinear time series prediction problems and have shown excellent performance in building energy consumption prediction [27].RNN is a type of artificial neural network, and its core idea is to introduce a recurrent structure so that the inputs at the current moment are not only correlated with the inputs of the previous moment but also correlated with the state of the previous moment. In this way, historical information can be encoded into the current state, thus capturing the long-term dependence of sequence data, effectively solving the long-term dependence problem of traditional neural networks in dealing with sequence data. RNN, as a powerful sequence modeling tool, also suffers from the long-term dependence problem, computational inefficiency, limitations in information transfer, sensitivity to sequence length, and difficulty in modeling complex nonlinear relationships, among other Disadvantages [28].

To overcome the drawbacks of RNN, LSTM neural network was proposed [29]. LSTM is a commonly used recurrent neural network that can effectively deal with sequential data, especially for long-term dependency problems. LSTM has a special memory unit that retains the information and passes it on to the next time step. LSTM is trained by using a gradient descent algorithm to learn how to recognize patterns and associations in an input sequence. Its ability to handle variable-length sequences and to learn dependencies with long-time delays has led to its widespread use in areas such as natural language processing, speech recognition, and time series prediction. However, LSTM has numerous hyperparameters, such as the number of neurons, learning rate, threshold, etc., and the hyperparameters have a great impact on its prediction results. Therefore, it has been an open research challenge to choose an intelligent algorithm to optimize its hyperparameters to improve the prediction accuracy [29].

1.6. Why beluga whale optimization algorithm?

A single data-driven model for building energy consumption prediction, dealing with data from multiple sources, can lead to problems of poor prediction accuracy or poor convergence. The method of combining advanced evolutionary algorithms with data-driven models provides a method for highly accurate and robust building energy consumption prediction models. In conclusion, the LSTM (IBWO-LSTM) prediction model based on improved beluga whale optimization (IBWO) proposed in this paper uses IBWO to optimize the hyperparameters of the LSTM neural network as a way to improve prediction accuracy.

In 2022, Zhong et al. proposed Beluga whale optimization (BWO), a novel population-based meta-heuristic algorithm inspired by beluga whale behavior, which involves beluga whale swimming, hunting, and dropping in the sea, to find the globally optimal solution. BWO has the advantages of high optimization accuracy and fast convergence speed, which can make it advantageous in a variety of real-world problems. BWO has the following points that can make it more effective than other meta-heuristic algorithms in obtaining the global optimal solution [30]:(1) BWO is an easy-to-implement technique as it is a derivative-free optimization technique. (2) BWO has better global convergence and it can well balance the exploration phase and exploitation phase. (3) BWO outperforms 15 meta-heuristic algorithms such as Genetic Algorithm, Particle Swarm Algorithm, Grey Wolf Optimization Algorithm, etc. in 23 out of 30 benchmarked test functions for single-peak, multi-peak and composite functions. (4) BWO is superior to other meta-heuristic algorithms in 67 out of 72 cases in the scalability analysis of benchmark functions of different dimensions.

1.7. Research gaps

By reading the extensive literature on building energy prediction, the research gaps identified are highlighted below.

(1) Artificial neural network is currently the most common prediction model in the building field. However it is lacking in time series prediction, whereas general building energy consumption data or loads are highly time dependent. Therefore it is limited in time series prediction and lacks the effectiveness and accuracy of time series prediction.
(2) According to the literature [17,30], it is known that the workload of determining the optimal hyperparameters of the prediction model by trial and error is great. To solve this problem, automatic searching of hyperparameters has been proposed, which can automatically search for hyperparameters of the prediction model.
(3) Although the literature mentions the use of evolutionary algorithms to determine the optimal hyperparameters of the prediction model, the algorithms used optimize with the goal of determining the optimal weight matrix of the LSTM or optimizing its hyperparameters, and it is not determined that optimizing several hyperparameters is optimal.

1.8. Research objectives of this study

The LSTM neural network, because of its structural characteristics, can predict time series and provide accurate and robust predictions of building energy consumption in practical problems. The main contributions of this paper are.

(1) Improving the convergence of simple and efficient BWO by employing chaotic mapping, nonlinear decreasing strategy, and nonlinear adjustment of the step factor proves to be an effective solution for identifying the optimal hyperparameter values of LSTM.

(2) The performance of LSTM is improved by determining the optimal hyperparameters of LSTM (number of neurons in the LSTM layer, learning rate, maximum number of iterations, and threshold) by employing an improved beluga whale optimization algorithm.
(3) In order to make up for the shortcomings of the previous articles that did not determine how many hyperparameters are optimal for optimization, this thesis uses the improved beluga whale optimization algorithm to optimize 2–4 parameters of the LSTM neural network respectively, to find out the best combination of optimization.

The rest of the paper is organized as follows: Section 2 introduces the basic theory of the proposed method, including the improvement method of BWO and the related concepts of LSTM; Section 3 focuses on the theory of energy consumption prediction models; Section 4 describes the relevant data sources of this study; and Section 5 describes the relevant experimental results: the conclusions and limitations obtained are listed in Section 6.

2. Method

2.1. Long-short term memory neural networks

LSTM neural network is a special type of Recurrent Neural Network (RNN). Fig. 1 shows a schematic diagram of an RNN neural network. The RNN network in the figure, x^t denotes the input of the current state accepted by each node, and h^{t-1} denotes the current hidden state value of the hidden layer accepted by each node [31]. Eqs. (1) and (2) are the basic calculations of the RNN neural network.

$$h^t = \delta\left(W_x^h * x^t + W_h^h * h^{t-1} + b^c\right) \tag{1}$$

$$y^t = \sigma\left(W_y^h * h^t + b^y\right) \tag{2}$$

Where: δ is the activation function; W_x^h, W_{h}^h and W_{y}^h denote the weight matrices of the input hidden layer, the hidden hidden layer and the hidden output layer, respectively, and b^h and b^y denote the hidden layer bias and output bias, respectively.

Compared to traditional RNNs, LSTM neural networks have more powerful memory capacity and can better handle long-term dependencies. In LSTM neural networks, the problem of gradient vanishing and exploding is solved by memory blocks in its structure [32]. The core idea of LSTM neural networks is to introduce a mechanism called "gates" to control the flow of information and forgetting. The memory block structure of the LSTM neural network is shown in Fig. 2.

At time step t, the input gate accepts two inputs at moment t, i.e., the current moment is the input data (x^t) and the output of the same memory cell obtained at the previous moment (h^{t-1}). Eps. (3)–(7) are the equations that control each gate in the LSTM neural network.

$$i^t = \sigma\left(* x^t + W_i^h * h^{t-1} + b^t\right) \tag{3}$$

$$g^t = Tanh\left(W_c^x * x^t + W_c^h * h^{t-1} + b^c\right) \tag{4}$$

The forgetting gate (f^t) indicates the amount of information to be removed from the memory block.

$$f^t = \sigma\left(W_f^x * x^t + W_f^h * h^{t-1} + b^f\right) \tag{5}$$

The update process of the storage unit is carried out by updating its value through the interaction of the state value of the previous moment (t-1) with the value of the forgotten gate.

$$c^t = f^t * c^{t-1} + i^t * g^t \tag{6}$$

The output gate (o^t) performs the output of the information, the memory cell is based on the activation function, the activation function has an effect on the value of the output and its determined that the output gate has to output that value.

$$o^t = \sigma\left(W_o^x * x^t + W_o^h * h^{t-1} + b^o\right) \tag{7}$$

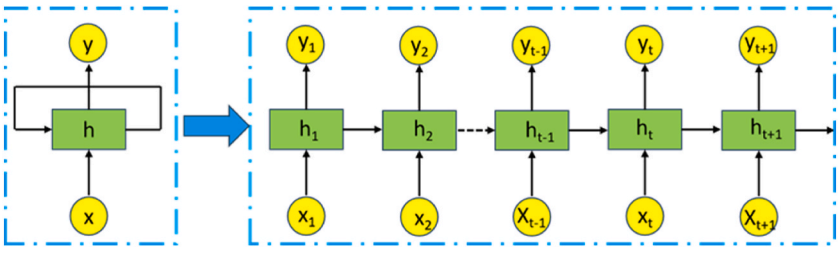


Fig. 1. Simple recurrent neural network.

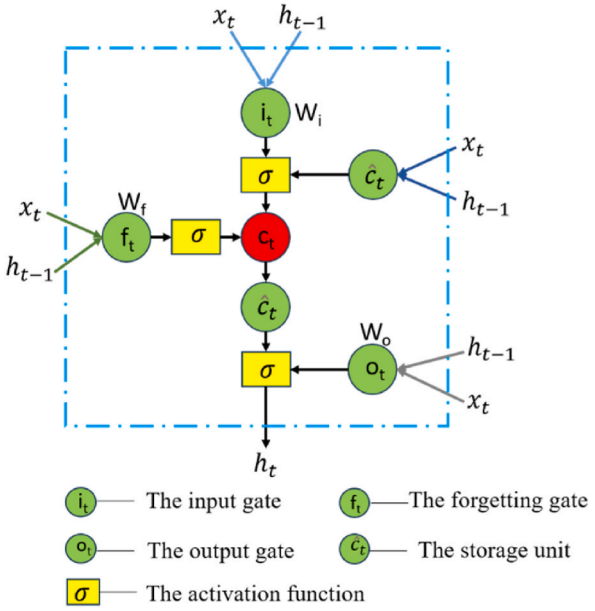


Fig. 2. Structure of the memory block of LSTM.

where W_i^x, W_c^x, W_f^x and W_o^x are the input weight matrices of each gate, and W_i^h, W_c^h and W_f^h are the recursive weight matrices of each gate, respectively.

The final output of the LSTM neural network is computed by calculating the output units of the hidden layer state values (h^t) and the overall output of the LSTM units (y^t). This is derived through equations (8) and (9).

$$h^t = o^t * Tanh(c^t) \tag{8}$$

$$y^t = \left(W_h^y * h^t + b^n\right) \tag{9}$$

where W_h^y is the weight matrix of the hidden layer output and b^n is the bias ($n \in (i,c,f,h)$).

Generally, LSTM uses the backpropagation algorithm of gradient descent (SGD) to update its parameters, but since the performance of SGD depends on the hyper-parameters of the LSTM such as the number of neurons in the hidden layer, the learning rate, the maximum number of iterations, and the threshold, an attempt has been made to determine the optimal values of these parameters as a means of improving the prediction accuracy of the LSTM in this paper.

2.2. Beluga whale optimization (BWO)

Metaheuristics is a class of algorithms based on advanced search and optimization ideas for solving complex optimization problems. These algorithms usually do not rely on the specific structure and constraints of the problem, but instead search for optimal or suboptimal solutions through randomization and iterative optimization. Metaheuristic algorithms have proven to be very effective methods in practical applications, especially for problems that are difficult to solve using traditional mathematical methods. There are many forms of meta-heuristic algorithms, among which the more common ones include genetic algorithm, simulated annealing, ant colony algorithm, particle swarm algorithm, and so on. The basic idea of all these algorithms is to search the solution space by randomization and iteration and to judge the quality of the current solution by an evaluation function.

Beluga Whale Optimization (BWO) is a population-based meta-heuristic algorithm proposed by Zhong et al. BWO is inspired by the behavior of beluga whales, which involves beluga whales swimming, hunting, and dropping into the sea. Similar to other metaheuristic algorithms, BWO contains an exploration phase and an exploitation phase. The exploration phase ensures global search capability in the design space through random selection of beluga whales, and the exploitation phase controls local search in the design space [30].

Since BWO is a population-based mechanism, each beluga is considered as a search agent in the algorithm, so each beluga is a candidate solution. It is updated in the optimization of the algorithm. The position matrix of the search agents is modelled as:

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,d} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,d} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,d} \end{bmatrix} \tag{10}$$

Where: n is the size of the beluga whale population, d is the dimension of the variable, and the corresponding matrix of fitness values is:

$$X=\begin{bmatrix} f(x_{1,1},x_{1,2},\ldots,x_{1,d}) \\ f(x_{2,1},x_{2,2},\ldots,x_{2,d}) \\ \vdots \\ f(x_{n,1},x_{n,2},\ldots,x_{n,d}) \end{bmatrix} \tag{11}$$

The BWO algorithm has an exploration phase and an exploitation phase, so the transfer to the exploitation phase depends on the equilibrium factor B_f, which is modelled as:

$$B_f = B_0(1 - T / 2T_{max}) \tag{12}$$

Where T_{max} is the maximum number of iterations, T is the current number of iterations and B_0 is a randomly varying quantity that varies between (0, 1) at each iteration. When $B_f > 0.5$, the algorithm is in the exploration phase, and when $B_f \leq 0.5$, the algorithm runs in the exploitation phase. When the number of iterations T increases, B_f also changes, from (0, 1) at the beginning to (0, 0.5), indicating that the probabilities of the exploratory and utilization phases change, with the probability of the exploratory phase decreasing.

The exploratory phase of BWO was established by considering the swimming behavior of beluga whales. Depending on their behavior under human attention, belugas can perform social behaviors in different postures, e.g. two pairs of belugas swimming closely together in a synchronized or mirrored manner [33]. Therefore, the position of the search agent is determined by the swimming of a pair of beluga whales and the position of the beluga whales is updated as follows:

$$\begin{cases} X_{i,j}^{T+1} = X_{i,p_j}^{T} + \left(X_{r,p_1}^{T} - X_{i,p_j}^{T}\right) + (1 + r_1)\sin(2\pi r_2), j = even \\ X_{i,j}^{T+1} = X_{i,p_j}^{T} + \left(X_{r,p_1}^{T} - X_{i,p_j}^{T}\right) + (1 + r_1)\cos(2\pi r_2), j = odd \end{cases} \tag{13}$$

where T is the number of iterations, $X_{i,j}^{T+1}$ denotes the new position of the ith beluga in the j-dimension, and $X_{i,j}^{T}$ is the position of the ith beluga in the j-dimension, where p_j ($j = 1,2,\cdots,d$) is a randomly chosen number in the d-dimension, and therefore, X_{r,p_1}^{T} and X_{i,p_j}^{T} denotes the position of the rth beluga (r denotes randomly chosen) with the position of the ith beluga, r_1 and r_2 denote random numbers within 0–1 used to augment the random arithmetic in the exploration phase, and $\sin(2\pi r_2)$ and $\cos(2\pi r_2)$ denote the fins facing the surface. Depending on the dimension chosen for odd and even numbers to update the position, it can reflect the synchronization of beluga whales in swimming or diving.

The utilization phase of the BWO was established by considering the feeding behavior of beluga whales. After observation, it was found that beluga whales cooperate to forage and move through the position of neighboring beluga whales. So the beluga whales forage by sharing the positional information with each other to consider the best candidate solution. In order to improve the convergence of the algorithm, Levy flight is introduced in the utilization phase of the BWO, which is mathematically modelled as:

$$X_i^{T+1} = r_3 X_{best}^{T} - r_4 X_i^{T} + C_1 \cdot L_F \cdot \left(X_r^{T} - X_i^{T}\right) \tag{14}$$

where T is the current number of iterations, X_i^{T} and X_i^{T+1} denote the current and new positions of the ith beluga whale, X_r^{T} denotes the position of a random beluga whale, X_{best}^{T} denotes the beluga whale at the best position, r_3 and r_4 denote the random numbers within 0–1, and $C_1 = 2r_4(1 - T/T_{max})$ denotes the random jumping strength, which is used to measure the strength of Levy's flight [33].

The Levy flight function L_F is calculated as:

$$L_F = 0.05 \times \frac{\mu \times \sigma}{|\nu|^{1/\beta}} \tag{15}$$

$$\sigma = \left(\frac{\Gamma(1+\beta) \times \sin(\pi\beta/2)}{\Gamma((1+\beta)/2) \times \beta \times 2^{(\beta-1)/2}}\right)^{1/\beta} \tag{16}$$

where μ and ν are random numbers obeying a normal distribution and β is a constant equal to 1.5.

As belugas migrate and hunt for food, they are threatened by creatures such as killer whales and polar bears, so some belugas do not survive and sink to the bottom of the sea, a phenomenon known as "whale fall". The beluga whales that fall to the bottom of the sea are food for a large number of organisms, and the remaining skeletons are broken down by bacteria or coral.

In order to model small changes in the population, subjective assumptions are made about the probability of whales dropping out of the population. It is assumed that beluga whales either move elsewhere or fall to the bottom of the ocean, and in order to ensure that the population size remains constant use the position of the beluga whales and the step size of the whales' descent to establish the updated position. The mathematical expression for this is:

$$X_i^{T+1} = r_5 X_i^{T} - r_6 X_r^{T} + r_7 X_{step} \tag{17}$$

Where r_5, r_6 and r_7 are random numbers within 0–1, and X_{step} is the step size of the whale's descent, which is denoted as:

$$X_{step} = (u_b - l_b)exp(-C_2 T / T_{max}) \tag{18}$$

Where C_2 is the step factor ($C_2 = 2W_f \times n$), the value of which is related to the probability of whale fall and the population size, u_b and l_b are the upper and lower bounds of the variables, respectively, and T is the current number of iterations. Therefore, the step factor is affected by the boundaries, the number of iterations and the population size.

The probability of a whale fall (W_f) is expressed as a linear model:

$$W_f = 0.1 - 0.05T/T_{max} \tag{19}$$

The above equation shows that the probability of a whale falling is 0.1 at the first iteration and 0.05 when the last iteration is performed, indicating that the probability of a beluga whale encountering danger decreases when it is close to food.

2.3. Improved beluga whale optimization (IBWO)

In the paper, three main improvements are used to improve the beluga optimization algorithm, namely, the initialization aspect (chaotic initialization), the parameter aspect (non-linear decreasing), and an improvement of the step factor for the whale falling phase. The convergence speed of the algorithm is enhanced and the two phases of the algorithm, global search, and local development, are balanced. The specific improvements are shown in Fig. 3.

(1) Cat Chaos Mapping

The Cat mapping is a two-dimensional invertible chaotic mapping with good population traversal uniformity and faster iteration speed when initializing the population [34]. Therefore, when the algorithm is initialized, the initial population generated using chaotic mapping has better diversity compared to the population randomly generated by the original algorithm and can be uniformly distributed in the solution space, which makes the algorithm optimization efficiency increase. Its dynamics equation is:

$$\begin{bmatrix} X_{n+1} \\ Y_{n+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} X_n \\ Y_n \end{bmatrix} mod\ 1 \tag{20}$$

Where X_n and Y_n denote the current population number, X_{n+1} and Y_{n+1} are the updated population number, and the mod is the modulus operation. The mapping structure is simple, with better traversal uniformity and faster iteration speed, and the chaotic sequences generated between [0, 1] are uniformly distributed.

(2) Non-linearly decreasing predation strategies

When the beluga whales make position updates, different search strategies are determined by the balancing factor B_{if}. In the exploration phase, more global search is needed to explore the whole solution space more comprehensively, and in the exploitation phase, each search agent is at the optimal solution, at which time the algorithm needs to carry out local exploitation, so that each beluga whale continues to search for a more optimal solution in their respective neighborhoods. Therefore, a nonlinear decreasing strategy is used, where B_{if} is dynamically reduced with the number of iterations, as a way to balance the algorithm's global search capability and local exploitation capability [35]. Its mathematical expression is:

$$B_{if} = a(1 - T/T_{max})^2 \tag{21}$$

where T and T_{max} are the current and maximum number of iterations, respectively, and a is a coefficient that takes the value of 1.

(3) Step factor (C_2) non-linear adjustment

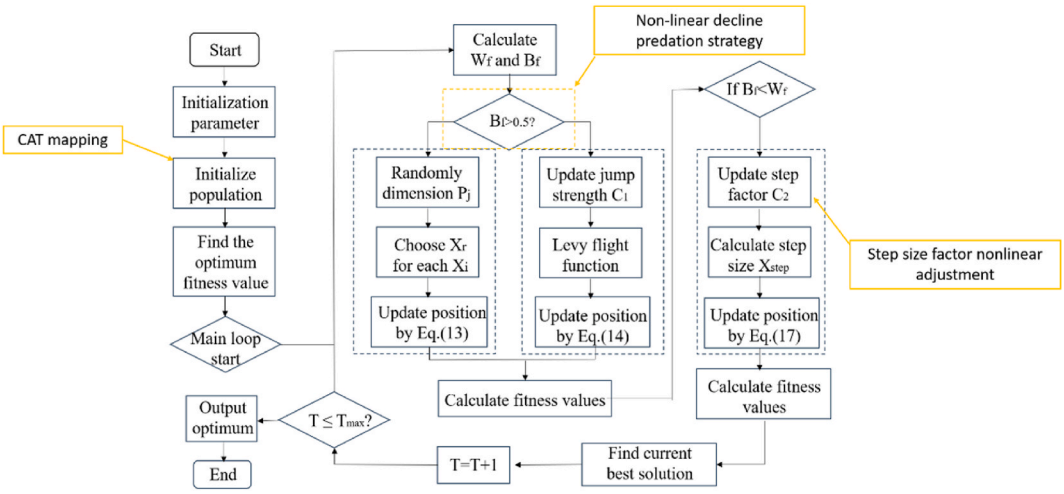


Fig. 3. Schematic diagram of the improved beluga optimization algorithm.

The step length (X_{step}) of the whale fall affects the position update of the whale fall stage, which is related to the step factor (C_2) as shown in equation (18).In the paper, C_2 is nonlinearly adjusted so that the probability of the whale falling when the beluga whale is close to the food shows a nonlinear change, so that the exploration stage and the utilization stage have an important effect on it. The mathematical expression is shown below:

$$W_{if} = fitness_{best} - (fitness_{best} - fitness_{worst})\left(\frac{T_{max} - T}{T_{max}}\right)^{2/3}. \tag{22}$$

Where $fitness_{best}$ and $fitness_{worst}$ are the best fitness and the worst fitness, respectively, and T_{max} and T are the maximum number of iterations and the current number of iterations. The above equation shows that the probability of whale fall is decreasing nonlinearly with the increase of the number of iterations.

3. Building energy Prediction modeling

The building energy prediction system proposed in this paper is composed of three main layers, namely (I) data acquisition and storage layer, (II) data preprocessing layer and (III) data analysis layer [13]. The system structure diagram of energy consumption prediction is shown in Fig. 4.

3.1. Data acquisition and storage layer

The data collection and storage layer play a vital role in energy consumption prediction. It is responsible for collecting energy consumption-related data from various data sources and storing it in a reliable and efficient database. Factors affecting energy consumption include climatic conditions, HVAC equipment, lighting systems, and the behavior of building occupants. In general buildings, human behavior, and equipment start and stop are irregular, but in commercial buildings, the operation of lighting and air conditioning is relatively stable throughout the day and is time-dependent. So LSTM is used as a prediction model for energy consumption prediction. Historical meteorological data include: outdoor dry bulb temperature, solar radiation intensity, relative humidity, and wind speed.

3.2. Data preprocessing layer

Raw data collected and stored by building equipment automation software usually includes all types of noise, anomalies, and missing data due to equipment failure or damage, sensor failure, or transmission errors. It is usually processed using sliding average filtering, sliding window, and linear interpolation techniques. The processed data is then normalized using the min-max normalization technique to address the stable convergence of model weights and biases. Finally, the normalized data is divided in 60:20:20 ratio for training, testing, and validation respectively.

3.3. Layering of data analysis

The data analysis layer uses the LSTM neural network to predict the energy consumption of the building, and the parameters of the LSTM neural network (number of neurons in the intermediate layer, learning rate, maximum number of iterations, and threshold) are optimized using an improved beluga whale optimization algorithm to enhance its prediction accuracy. The performance of IBWO-LSTM is evaluated using test set data.

3.4. IBWO-LSTM: a data-driven predictive model for building energy consumption

The goal of IBWO is to determine the optimal hyperparameters of the LSTM neural network, i.e., the number of neurons in the middle layer, the learning rate, the maximum number of iterations, and the threshold value. The overall process of IBWO-LSTM can be represented by four phases, which are (I) Setting of the initial values of the parameters, i.e., generating the initial population of Cat chaos (II) Optimization of the hyperparameters (III) Iteration of algorithmic updating (IV) Performance of the model Evaluation. The specific process is as follows.

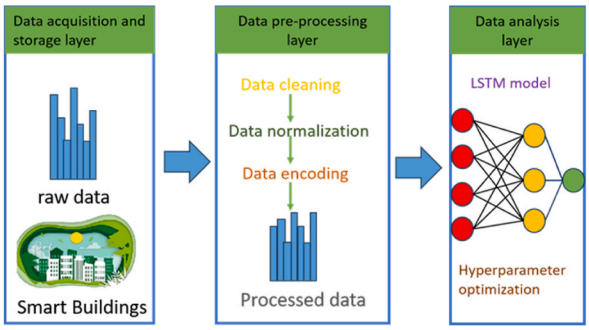


Fig. 4. Architecture of building energy Prediction system.

Step 1 Collect historical energy consumption data, divide it into training set, testing set and validation set, and then normalize it.
Step 2 Optimization of hyperparameters of LSTM neural network using IBWO algorithm.
Step 2.1 Constructing Predictive Models for LSTM Neural Networks.
Step 2.2 Set the parameters of BWO, which include the population size and maximum number of iterations. Initialize the beluga whale population using Eq. (20) to generate the Cat chaotic mapping.
Step 2.3 Balance the two phases of global search and local exploitation according to Eq. (21) so that the exploration phase performs more global search and finds a more optimal solution during the exploitation phase.
Step 2.4 In the whale falling stage, C_2 is nonlinearly adjusted according to Eq. (22), so that the probability of the beluga whale falling when approaching the food presents a nonlinear change, and the algorithm optimization seeking ability is enhanced.
Step 3 The algorithm is iteratively updated until the hyperparameter values are output to the LSTM neural network when the conditions are satisfied.
Step 4 The test set is put into the energy prediction model to predict the energy consumption of the building, and the predicted value of energy consumption is obtained, and the prediction results are evaluated.

The structure of the proposed IBWO-LSTM model is shown in Fig. 5.

4. Case study

4.1. Project introduction

The proposed energy prediction system is tested in an office building in the western city of xi ' a to evaluate its performance. The office building has 44 floors above ground and 3 floors below ground, with a total floor area of 300,000 m^2 and a site area of 30,000 m^2. The wall structure of the building is a concrete shear wall with a body factor of 0.8. The building is deployed with environmental detection sensors and energy consumption collection systems that are capable of collecting real-time outdoor temperature, humidity, and energy consumption data. The data types are shown in Fig. 6.

4.2. Data sources

The data collected is energy consumption data from May 1, 2022 to August 31, 2022, energy consumption data is collected on an hourly granularity from 8 a.m. to 22 p.m. daily. The data includes air conditioning power consumption, special power consumption, power consumption, and lighting socket consumption. Because the data used in this paper were collected during the summer period, the air conditioning power consumption is mainly the power generated by the central air conditioning cooling in the shopping mall; the special power consumption mainly includes the power generated by the water supply and drainage system and the weak power room in the shopping mall; the power consumption is mainly the power generated by the elevators, escalators, and moving walks; the lighting outlet power consumption is mainly the power generated by the lighting equipment in the shopping mall, including the LED

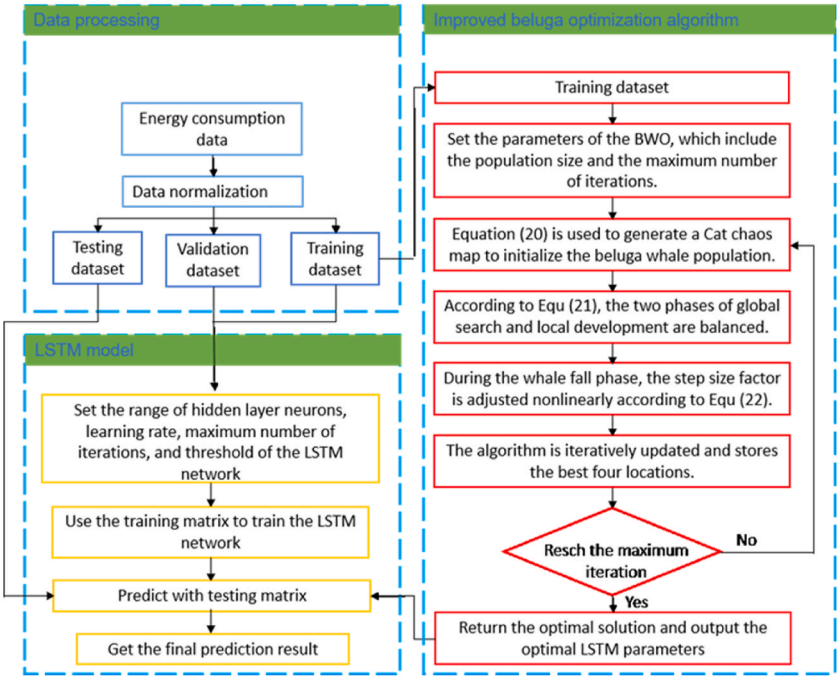


Fig. 5. IBWO-LSTM: Flowchart of building energy consumption Prediction model.

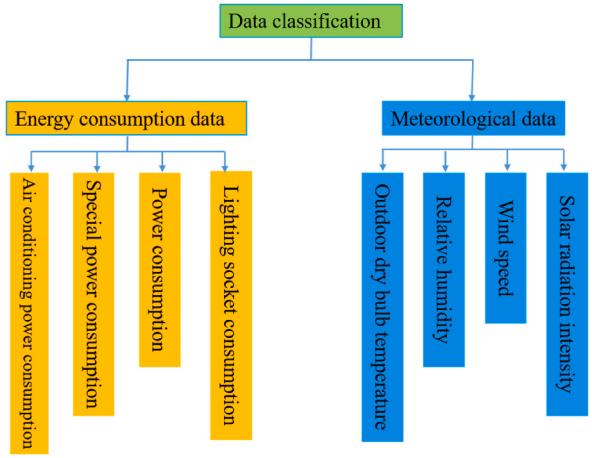


Fig. 6. Energy consumption data, meteorological data classification chart.

lamps, fluorescent lamps, and the sockets in the mall are used to supply various electronic devices and household appliances. Fig. 7 gives the four types of electricity loads from 8 a.m. to 22 p.m. on 31 August for the forecast period [17,36].

Meteorological data include outdoor dry bulb temperature, relative humidity, wind speed, and solar radiation intensity. Fig. 8 shows the outdoor meteorological data information of the forecast from 26 August to 30 August, and the outdoor temperature varies in a more uniform range from 19.6 °C to 30.3 °C. In addition, the remaining three meteorological data are affected by weather conditions and uncertainty factors. In this paper, outdoor dry bulb temperature, relative humidity, wind speed, solar radiation intensity, time of

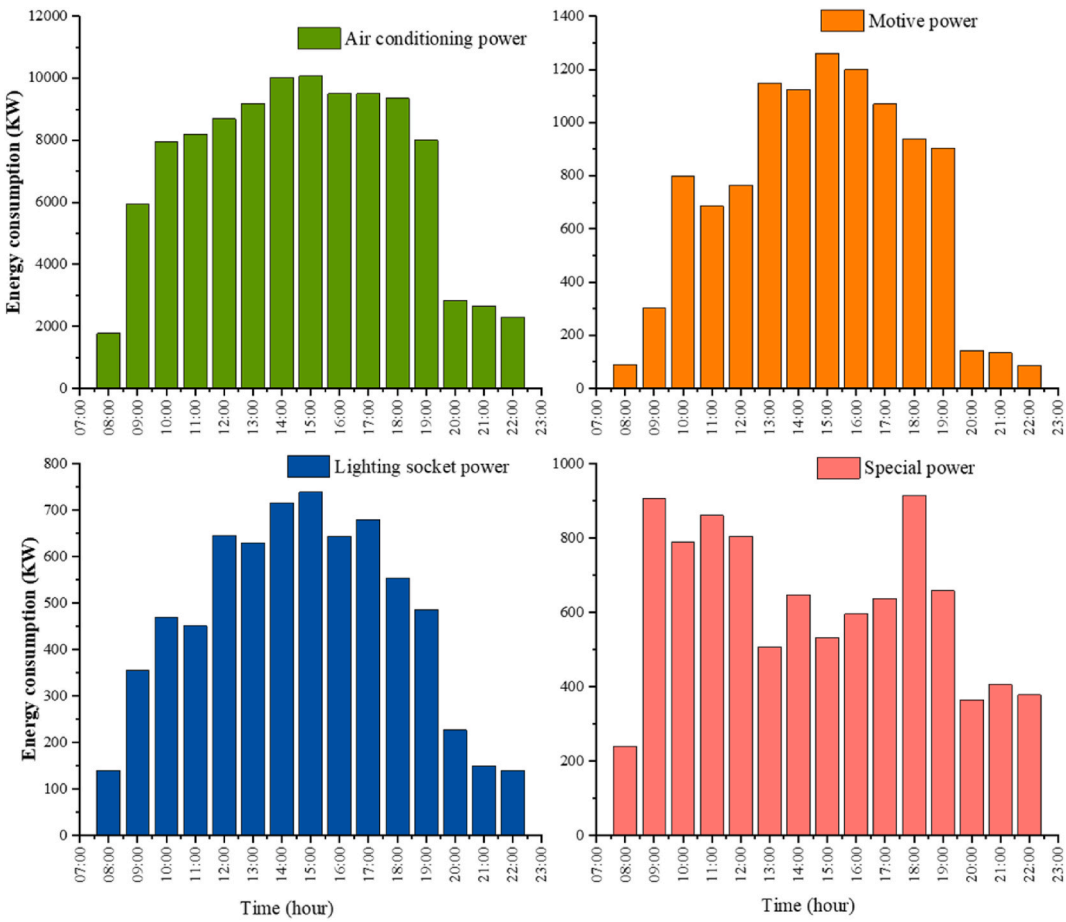


Fig. 7. Four categories of electricity load on 31 August.

day, and historical energy consumption data are used as inputs to the prediction model.

The data are collected hourly from 8:00 to 22:00 every day, as shown in Table 2. The prediction model is divided into two parts: the first part is the construction of the LSTM prediction model, which is trained and validated using a total of 1360 sets of data from May 1 to August 1. The LSTM model is tested using a total of 450 sets of data from August 2 to August 30, which is the same as the LSTM prediction model. The LSTM model is tested using a total of 450 sets of data from 1 August to 30 August. The second part is the prediction of the results, using the number of 31 August for prediction and then evaluating the model.

5. Results and discussion

This part mainly deals with demonstrating the accuracy of IBWO-LSTM for building energy consumption prediction for the considered data-driven model by performing a comparative analysis of the prediction of energy consumption in office buildings. The main objective of this research work is to design an efficient, robust building energy consumption prediction model with high prediction accuracy and low complexity. IBWO-LSTM is an improved version of LSTM neural network, which combines the improved beluga whale optimization algorithm with LSTM to predict the energy consumption of an office building in xi, an. The parameter settings of IBWO-LSTM are given in Table 3.

In this paper, two hyperparameters (Number of hidden units and Learning rate), three hyperparameters (Number of hidden units, Learning rate and Number of iterations) and four hyperparameters (Number of hidden units, Learning rate, Number of iterations and Threshold) of LSTM neural network are optimized and then compared with BWO-LSTM, PSO-LSTM, PSO-LSTM, PSO-LSTM and Threshold. Number of hidden units, Learning rate and Number of iterations and Threshold) are optimized and the one with better optimization results is selected and compared with four energy prediction models, namely, BWO-LSTM, PSO-LSTM, GWO-LSTM and TSA-LSTM, to demonstrate that IBWO-LSTM is the most effective method for predicting energy consumption. demonstrate the effectiveness of the IBWO-LSTM model. All the runs of the results in the paper are implemented in Matlab 2019a and the average values (Relative Error, MAPE, MAE and RMSE) obtained from each model after 10 runs are used to comparatively analyze the performance of the IBWO-LSTM model. The mathematical model for evaluating the parameters is:

$$Relative\ Error = \frac{|\widehat{y_i} - y_i|}{y_i} \tag{23}$$

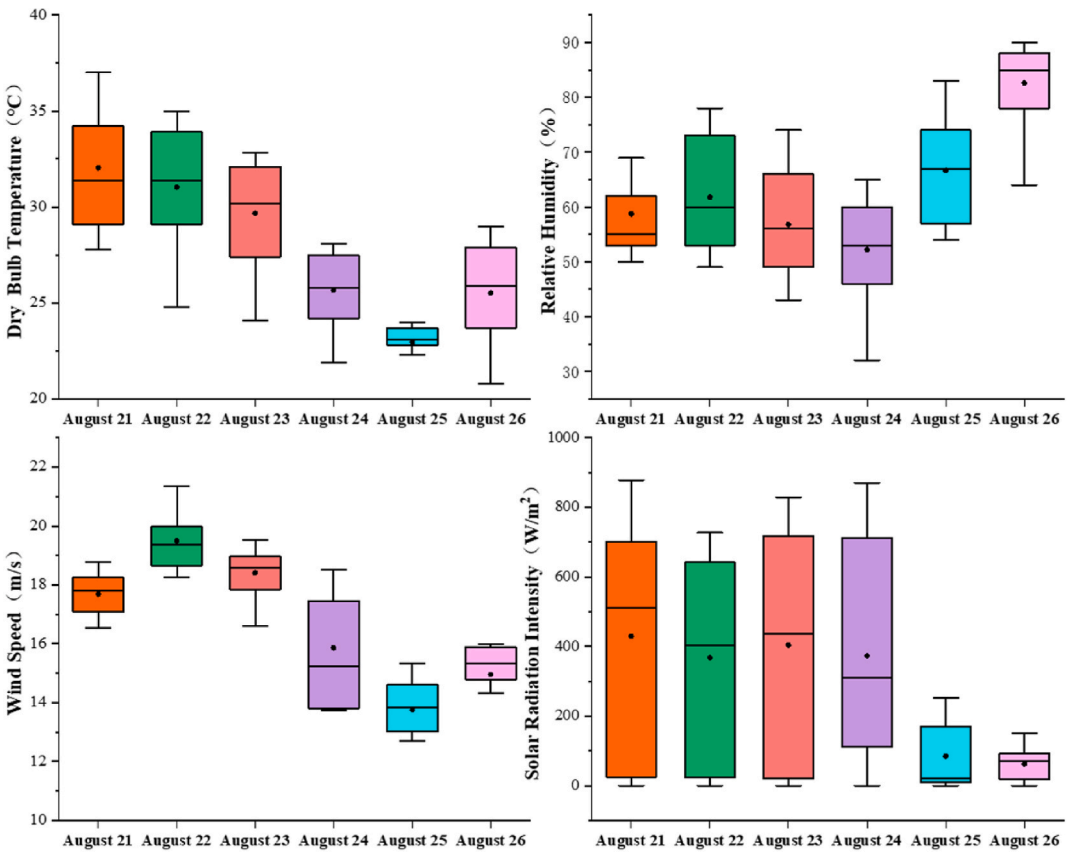


Fig. 8. Schematic representation of meteorological data.

Table 2
Predictive modeling dataset segmentation.

Name	Number of data sets	Time period
Training set	1360	5.1–8.1
Test set	450	8.2–8.30
Forecast day	1	8.31

Table 3
Parameter setting: LBWO-LSTM.

S.no	Parameter	Range
1	Learning rate	[0.001,0.1]
2	Number of hidden units	[10,300]
3	Number of iterations	[50,100]
4	Threshold	[0,1]
5	Number of populations	6
6	Dimension	4

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{|\widehat{y}_i - y_i|}{y_i}\right) \times 100\% \tag{24}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|\widehat{y}_i - y_i| \tag{25}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\widehat{y}_i - y_i)^2} \tag{26}$$

$$CV = \frac{\sqrt{\frac{\sum_{i=1}^{N}(\widehat{y}_i - y_i)^2}{N}}}{\overline{y_i}} \times 100\% \tag{27}$$

Where: y_i denotes the actual value, $\widehat{y}_i$ denotes the predicted value and $\overline{y_i}$ denotes the average value.

5.1. Building energy consumption prediction modeling results

The data ranges for the optimization based on two, three and four parameters are given in Table 4. The optimization is first performed for the number of neurons in the hidden layer and the learning rate, and the range of the optimization parameters as well as the values of the remaining two parameters are determined through the literature [13,21]. Then the optimization is performed for the number of neurons in the hidden layer, the learning rate and the maximum number of iterations, the range of the optimization parameters as well as the value of the threshold are determined through the literature [24], and finally all the four parameters are optimized, and then the accuracy of the three results are compared and analyzed.

Fig. 9 gives the errors of the LSTM neural network optimized based on two, three and four parameters and the model performance are optimal when optimized for four hyperparameters after analytical evaluation. Fig. 10 shows that the prediction accuracy increases with the number of optimized parameters when the LSTM neural network hyperparameters are optimized using the improved BWO.

Table 5 gives the results of hyperparameter selection for three models based on IBWO-optimized LSTM with two, three, and four parameters as well as for four models of BWO-LSTM, PSO-LSTM, GWO-LSTM and TSA-LSTM.

The results of seven different prediction models for predicting the energy consumption of office buildings are given in Fig. 11. It contains three models with IBWO optimized LSTM parameters and four models with BWO-LSTM, PSO-LSTM, GWO-LSTM and TSA-LSTM.

The relative errors of the seven different prediction models are given in Fig. 12. When the prediction model predicts the building energy consumption for a day, the relative error of the prediction model is larger when predicting these two time periods because of the

Table 4
Parameter ranges based on 2-, 3- and 4-parameter optimization.

Forecasting scenario	IBWO-LSTM-2	IBWO-LSTM-3	LBWO-LSTM-4
Number of hidden units	[10,100]	[10,100]	[10,100]
Learning rate	[0.001,0.1]	[0.001,0.1]	[0.001,0.1]
Number of iterations	50	[50,100]	[50,100]
Threshold	1	1	[0,1]

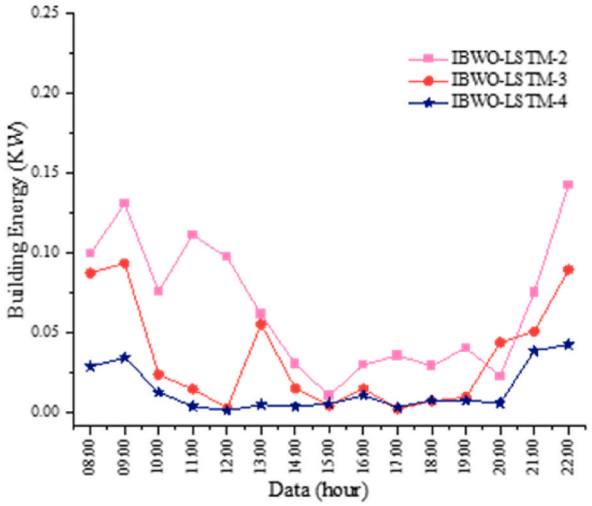


Fig. 9. Error analysis based on 2, 3 and 4 parameter optimization.

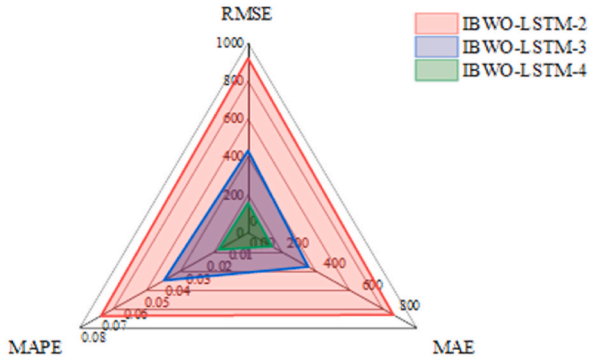


Fig. 10. Performance comparison based on 2, 3 and 4 parameter optimization.

Table 5
Model hyperparameter selection results.

Forecasting scenario	IBWO-LSTM	BWO-LSTM	PSO-LSTM	GWO-LSTM	TSA-LSTM
Number of hidden units	[10,300]	[10,300]	[10,300]	[10,300]	[10,300]
Learning rate	[0.001,0.1]	[0.001,0.1]	[0.001,0.1]	[0.001,0.1]	[0.001,0.1]
Number of iterations	[50,100]	[50,100]	[50,100]	[50,100]	[50,100]
Threshold	[0,1]	[0,1]	[0,1]	[0,1]	[0,1]
Population size	20	20	20	20	20
Population iterations	30	30	30	30	30

large changes in the building energy consumption data in the morning as well as in the evening of a day, and the relative error is smaller in other time periods, the right graph in Fig. 12 is a zoomed-in view of the left graph with the relative error for the morning as well as the evening time periods removed. From the figure, it can be seen that the BWO-LSTM has the largest relative error, which ranges from 4 % to 68 %, followed by GWO-LSTM and IBWO optimizing the hyperparameters of the 2 LSTMs, whose relative errors range from 3 % to 31 %. The smallest relative error is for IBWO optimization of 4 LSTM hyperparameters whose relative error is between 1 and 4%. Therefore, by improving BWO so that it optimizes the LSTM hyperparameters improves the prediction accuracy of the model as well as optimizing the parameters, optimizing 4 parameters has higher prediction accuracy than optimizing 2 and 3 parameters. The specific performance evaluation is shown in Table 6.

When using IBWO to optimize the four hyperparameters of LSTM, although the prediction accuracy is better than several other models, the relative error increases when the data fluctuates a lot, as shown in Fig. 12, in the two time periods of 8:00–10:00 and 20:00–22:00, because of the large data changes, the relative error of the model prediction also increases, which is the shortcoming of this model, and the future research can solve this problem by adding more datasets and optimizing more hyperparameters.

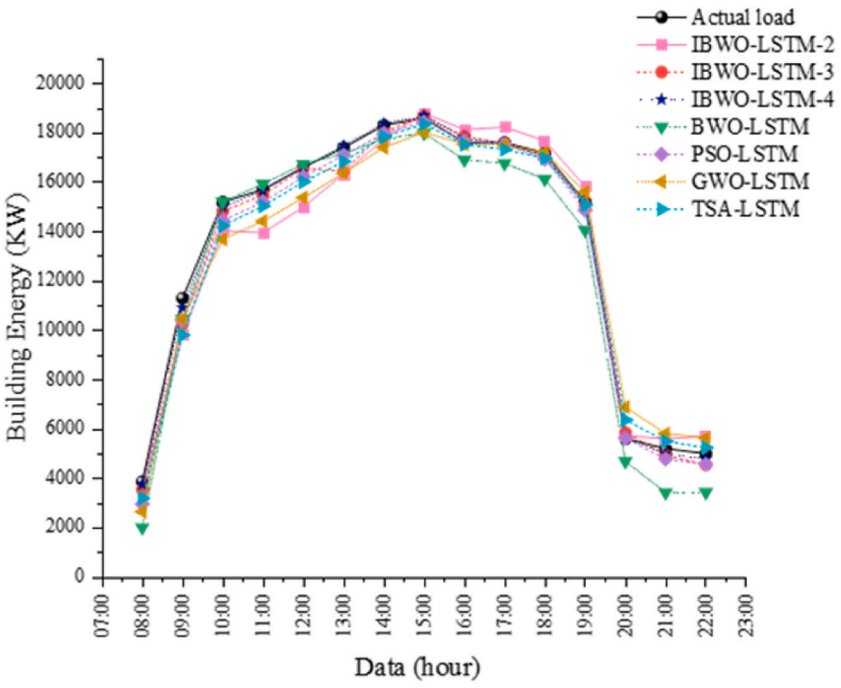


Fig. 11. Predictions from different forecasting models.

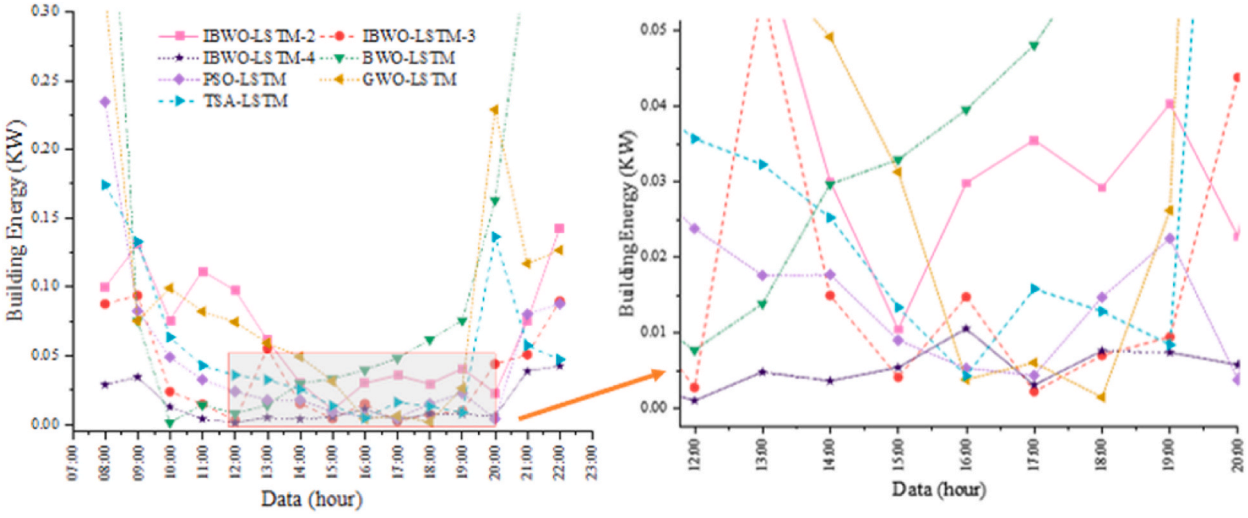


Fig. 12. Relative errors of different models in predicting building energy consumption.

Table 6
Error analysis between different forecasting models.

Forecasting methods	MAE	MAPE	RMSE	CV
BWO-LSTM	830.71	0.1232	1001.42	0.075
GWO-LSTM	779.62	0.0863	908.61	0.068
TSA-LSTM	510.13	0.0533	625.63	0.047
PSO-LSTM	393.61	0.0455	477.70	0.036
IBWO-LSTM-2	778.74	0.0660	921.34	0.070
IBWO-LSTM-3	322.13	0.0339	434.01	0.033
IBWO-LSTM-4	128.28	0.0138	158.36	0.012

6. Conclusions

In this study, a building energy prediction model is proposed, which uses IBWO and LSTM for accurate building energy prediction. The BWO employs Cat chaotic mapping, which distributes the initial population uniformly in the solution space to increase the population diversity, and a nonlinear decreasing strategy, which balances the global search with the local search to increase the algorithm's ability to find the optimal value. The hyperparameters (number of neurons in the hidden layer, learning rate, maximum number of iterations and threshold) of the LSTM neural network are optimized using improved BWO to find the optimal values of the parameters. The energy consumption of an office building in the western city of Xi'A is used as an example to predict the energy consumption from 8:00 a.m. to 10:00 p.m. in a day (31 August). During the experiments, the effect of optimizing 2 to 4 hyperparameters of the LSTM neural network using IBWO on the prediction accuracy was analyzed. The results show that the best performance of the prediction model is achieved when optimizing 4 hyperparameters, which reduces the RMSE by 11 % and the CV by 6.3 % compared to the BWO before improvement. In addition, the performance of IBWO-LSTM was verified in advanced energy consumption prediction models in statistics such as MAE, MAPE, RMSE.

This research can provide new insights on how to optimize and effectively employ computational methods. In other words, the hyperparameters of machine learning (SVM, ANN, CNN, etc.) can be optimized by different optimization methods (e.g., PSO, BWO, GWO, etc.) and used to enhance the prediction accuracy and robustness of machine learning. To better enhance the performance of predictive models, more hyperparameters of machine models can be optimized as well as new evolutionary algorithms can be proposed.

There are three main limitations that may exist in IBWO-LSTM: (I) The effect of climate on the energy consumption values in the energy consumption data is not analyzed. (II) There is no pre-processing technique for faults such as sensor drift during data collection. (III) There is no prediction of the energy consumption of the building in the medium and long term.

CRediT authorship contribution statement

Zengxi Feng: Writing – review & editing, Supervision, Resources. **Jianhu An:** Writing – original draft, Validation, Software, Methodology. **Mingyue Han:** Methodology, Data curation. **Xiuming Ji:** Validation, Resources. **Xian Zhang:** Conceptualization. **Chang Wang:** Software, Methodology. **Xuefeng Liu:** Formal analysis, Conceptualization. **Limin Kang:** Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Abbreviations	Description
Introduction	
ANN	Artificial neural network
CNN	Convolutional neural network
RNN	Recurrent neural network
PNN	Probabilistic neural network
ARMA	Autoregressive moving average model
ARIMA	Auto regressive integrated moving average
BWO	Beluga whale optimization
LSTM	Long short term memory
IBWO-LSTM	Improved beluga whale optimization based LSTM
MSE	Mean square error
RMSE	Root mean square
MAE	Mean absolute error
MAPE	Mean absolute percentage error
Long Short Term Memory	
x^t	Input value of the current time stamp
h^t	Hidden state value of the current time stamp

(*continued on next page*)

(*continued*)

Abbreviations	Description
y^t	Input value of the current time stamp
W_x^h, W_h^h and W_y^h	Weight matrix of the input-hidden layer, hidden-hidden layer, hidden-output layer respectively
b^h and b^y	Hidden and output bias
i^t	Input gate
g^t	Update gate
f^t	Forget gate
o^t	Output gate
W_i^x, W_c^x, W_f^x and W_o^x	Input weight matrices
b^n	Bias
Beluga Whale Optimization	
T_{max}	Total number of iterations
T	Current iteration
X*	Population location

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